MS&E 226 — Midterm 1 Practice Questions (Generated Set 2)

16 questions spanning the entire Midterm 1 syllabus (Lectures 2–9 plus Problem Set 3 topics: weighted least squares, quantile regression, missing-data imputation, log loss, and decision trees). Difficulty is calibrated against the office-hours Problem Set 3 practice quiz rather than Generated Set 1 – expect concrete population models, two-statement "(i)/(ii), which is correct" evaluations, and transfer/edge-case reasoning rather than pure recall. Multiple choice, one correct answer among (a)–(d); "(e)" lets you skip a question.

Question 1

An online retailer fits OLS to model delivery cost as a function of shipping method, a categorical feature with three levels: Ground (the baseline category), Air, and Express. The model is `cost ~ 1 + shipping`, and R reports: intercept $= 12$, `shippingAir` $= 16$, `shippingExpress` $= 33$ (all in dollars). What does the coefficient on `shippingExpress` represent?

(a) The difference between the mean delivery cost for Express shipments and the mean delivery cost for Ground (baseline) shipments.

(b) The mean delivery cost for Express shipments, in dollars.

(c) The mean delivery cost for Express shipments minus the overall average delivery cost across all three shipping methods.

(d) The change in delivery cost associated with a one-unit increase in "amount of express shipping."

(e) I choose not to answer this question.

Question 2

You include two highly correlated (but not perfectly collinear) features – years of experience and age – in an OLS model predicting salary. Evaluate the following statements:

(i) The model's predictions (and R^2) are reliable despite the high correlation between the two features.

(ii) The individual coefficients on experience and age can each be interpreted as that feature's "pure," independent effect on salary, unaffected by the correlation between them.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Question 3

You evaluate several candidate models on a validation set and select the model with the lowest validation RMSE as the "winner." You then evaluate that winner on a separate test set that played no role in selecting the winner. Evaluate:

(i) The winner's validation RMSE is an unbiased estimate of its true generalization error.

(ii) The winner's test RMSE is an unbiased estimate of its true generalization error.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Question 4

You fit a ridge regression using two features – income (ranging from \$10,000 to \$500,000) and a 0/1 indicator for `has_degree` – without standardizing either feature first. What is the consequence of skipping standardization here?

(a) Because the penalty $\lambda \sum_j \hat{\beta}_j^2$ is applied to the raw coefficient values, and income needs a tiny coefficient (since it's measured in large dollar units) while `has_degree` needs a much larger coefficient for comparable predictive impact, the penalty shrinks the two features' coefficients asymmetrically even if the features are equally predictive.

(b) There's no real issue – ridge regression is scale-invariant by construction, so standardization is purely a convenience, not a requirement.

(c) Skipping standardization makes the fit biased even at $\lambda = 0$, since the unpenalized OLS solution itself depends on feature scale.

(d) Skipping standardization only matters for lasso's sparsity property; ridge's shrinkage behavior is unaffected by feature scale.

(e) I choose not to answer this question.

Question 5

From a dataset with 500 candidate features, you compute each feature's correlation with the outcome using the **entire** dataset, keep the 10 most correlated features, and then run 10-fold CV using only those 10 features to estimate the generalization error of the final model (which will be fit on all the data using those 10 features). Is the resulting Err_{CV} trustworthy?

(a) No – Err_{CV} will typically **underestimate** the true generalization error, because the 10 features were chosen using information from every fold, including each fold when it is later held out, so the held-out folds aren't fully out-of-sample with respect to the feature-selection step.

(b) No – Err_{CV} will typically **overestimate** the true generalization error, because using only 10 of 500 features makes the model simpler and therefore harder to fit well.

(c) Yes, this is fine, because the outcome variable itself is never used inside any fold's model-**fitting** step – only in the earlier screening step.

(d) Yes, this is fine, as long as $K \geq 5$.

(e) I choose not to answer this question.

Question 6

Compare $K = 5$ vs. $K = n$ (leave-one-out) cross-validation for estimating the generalization error of a model that will ultimately be trained on the **entire** dataset. Evaluate:

(i) Using $K = 5$ instead of $K = n$ tends to **overestimate** the true generalization error of the final (full-data) model, since each fold's model is trained on less data than the final model will be.

(ii) Using $K = n$ always gives a strictly more precise (lower variance) estimate of generalization error than $K = 5$, simply because it uses more folds.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Question 7

A screening test is applied to $n = 2000$ patients. Of the 40 patients who actually have the condition, the test flags 30 correctly (and misses 10). Of the 1960 patients who don't have the condition, the test incorrectly flags 40 as positive. What is the **false discovery proportion** (FDP) of this test – the fraction of its positive predictions that are actually wrong?

(a) $\frac{40}{70} \approx 0.57$

(b) $\frac{10}{40} = 0.25$

(c) $\frac{40}{1960} \approx 0.02$

(d) $\frac{40}{2000} = 0.02$

(e) I choose not to answer this question.

Question 8

Classifier A has AUC $= 0.80$; Classifier B has AUC $= 0.75$, but their ROC curves cross: B's curve lies **above** A's for false positive rates below 0.1, and below A's curve everywhere else. A hospital screening for a dangerous condition is only willing to tolerate a false positive rate of at most 5% (i.e., it will operate strictly within that crossing region). Which classifier should it prefer, and why?

(a) Classifier B – at the hospital's actual operating point (FPR $\leq 5\%$, inside the region where B's curve lies above A's), B achieves a higher true positive rate there, even though A has the higher **overall** AUC. AUC summarizes performance averaged across all thresholds, not specifically at the one operating point that matters here.

(b) Classifier A, because it has the higher AUC, and AUC is always the right criterion for choosing between classifiers regardless of context.

(c) It doesn't matter which one they choose, since both classifiers achieve the same accuracy at any fixed threshold.

(d) Classifier A, because a crossing ROC curve means Classifier B's results are unreliable.

(e) I choose not to answer this question.

Question 9

Which of the following is **not** one of the three steps in the general maximum likelihood estimation (MLE) recipe presented in lecture?

(a) Assume a distributional form for how the data was generated.

(b) Compute (write down) the likelihood of the observed data under that assumed distribution.

(c) Choose the parameter values that maximize that likelihood.

(d) Split the data into training and validation sets to prevent overfitting.

(e) I choose not to answer this question.

Question 10

Which of the following is correct?

(a) Logistic regression's negative log-likelihood (log loss) is convex in the coefficients, so any local minimum found by an iterative optimizer is guaranteed to be the global minimum; this guarantee does **not** carry over to neural networks, whose loss surfaces are generally non-convex.

(b) Both logistic regression and neural network training are convex optimization problems, since both minimize a loss function using gradient-based methods.

(c) Logistic regression's loss is non-convex, which is exactly why iteratively reweighted least squares is needed instead of plain gradient descent.

(d) Convexity is a property of the dataset, not the loss function, so whether logistic regression's optimization is convex depends on whether the data happens to be linearly separable.

(e) I choose not to answer this question.

Question 11

You fit ridge regression on **standardized** features, so that as $\lambda \to \infty$, every feature coefficient is driven to exactly zero and the model's prediction converges to just the intercept (which, for standardized features, equals $\overline{Y}$). Assuming the true population model is **not** simply a constant, what happens to the model's bias and variance as $\lambda \to \infty$?

(a) Both converge to the bias and variance of the constant-mean predictor $\overline{Y}$: variance shrinks to the (generally small but nonzero) variance of the sample mean, while bias converges to a fixed, generally **nonzero** value – the gap between the true conditional mean and the unconditional population mean. Ridge trades away its features' predictive content for stability, landing at a high-bias, low-variance extreme.

(b) Both bias and variance converge to zero, since $\lambda \to \infty$ makes the model maximally regularized and therefore maximally accurate.

(c) Variance converges to zero (the model no longer depends on any feature), but bias stays exactly zero throughout, just as it was for OLS under (A1)–(A3), since ridge never introduces bias.

(d) Both bias and variance increase without bound as $\lambda \to \infty$, since an infinitely large penalty destabilizes the fitting procedure.

(e) I choose not to answer this question.

Question 12

Suppose the population model is $Y = 4X + \varepsilon$, where $X > 0$, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathrm{Var}(\varepsilon \mid X) = \frac{16}{X^2}$ (the error variance **shrinks** as X grows). You have a sample in which X takes many different values, and you estimate the coefficient on X (no intercept) two ways: by OLS, and by WLS with weights $w_i = X_i^2$. Evaluate:

(i) The OLS estimate of the coefficient is unbiased.

(ii) The WLS estimate of the coefficient has lower variance than the OLS estimate.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements (i) and (ii) are correct.

(d) Neither statement (i) nor (ii) is correct.

(e) I choose not to answer this question.

Question 13

Suppose the population model is $Y = 10 + 0.5X + X\varepsilon$, where $X > 0$ and ε is normally distributed with mean zero and variance 4, independent of X (so the spread of Y around $10 + 0.5X$ grows with X). Consider the population conditional quantiles of Y given X – the true lines that quantile regression with $\tau = 0.2$ and $\tau = 0.7$ are trying to estimate. Which of the following is correct?

(a) The $\tau = 0.2$ and $\tau = 0.7$ lines are parallel.

(b) The $\tau = 0.2$ line has a larger slope than the $\tau = 0.7$ line.

(c) The $\tau = 0.7$ line has a larger slope than the $\tau = 0.2$ line.

(d) Both lines are the same as the conditional mean line $10 + 0.5X$.

(e) I choose not to answer this question.

Question 14

When using k-nearest-neighbors imputation to fill in missing feature values, which of the following correctly describes the tradeoff in choosing k?

(a) Small k makes the imputed value highly sensitive to the noise in just a few nearby observations (high variance); large k averages over less locally-relevant observations, smoothing out genuine local structure (introducing bias) – directly analogous to the bias-variance tradeoff in using k-NN for prediction.

(b) Small k always introduces more bias, since it ignores most of the dataset; large k always has the lowest variance, since it uses the most available data.

(c) The choice of k for imputation has no connection to the bias-variance tradeoff discussed for prediction, since imputation is a preprocessing step, not a predictive model.

(d) Larger k is always preferable for imputation, since it uses more information and is therefore less likely to be wrong on any single missing value.

(e) I choose not to answer this question.

Question 15

You fit regression trees on the same training set using the same deterministic tree-building procedure, with no complexity penalty, varying only the maximum depth, and use 5-fold cross-validation to estimate generalization error at each depth. Evaluate:

(i) Shallow trees (small maximum depth) tend to have higher bias than deeper trees.

(ii) Cross-validation RMSE is minimized at the deepest tree considered, since deeper trees always fit the training data at least as well.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Question 16

Let $\hat{p}$ be the fraction of "successes" in a sample of n i.i.d. Bernoulli(p) trials (e.g., the fraction of surveyed voters who support a candidate), used as an estimator of the true population proportion p. Using the estimator bias-variance framework ($\text{MSE} = \text{Bias}^2 + \text{Variance}$, no irreducible-error term), which of the following is correct about $\hat{p}$ as an estimator of p?

(a) $\hat{p}$ is unbiased for p ($\mathbb{E}[\hat{p}] = p$), and its variance is $\frac{p(1-p)}{n}$, which shrinks as n grows – so $\text{MSE} = 0 + \frac{p(1-p)}{n} \to 0$ as $n \to \infty$.

(b) $\hat{p}$ is unbiased, but since estimating a proportion involves averaging random (0/1) outcomes, MSE must also include an irreducible error term reflecting the randomness of each individual trial.

(c) $\hat{p}$ is biased toward 0.5, since averaging binary outcomes pulls the estimate toward the middle of its possible range.

(d) The variance of $\hat{p}$ is $p(1-p)$ and does not depend on n, since variance is a population quantity, not a sample quantity.

(e) I choose not to answer this question.